Computer Science & IT

C Programming

Functions & storage classes

Lecture - 01



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Recap of Previous Lecture



- Topic — Nested for loop
- Topic — Some program
- Topic — Break , continue
- Topic — While and do while
- Topic

Slide

Topics to be Covered



Storage class: function

Topic	Function , prototype , parameter & arguments
Topic	Function Declaration
Topic	Function definition
Topic	Function call
Topic	

Slide



do-while loop

do while loop : exit controlled loop

A loop in which condition is tested after execution of body of loop.



do-while loop

```
int i=1
do{
    printf("I am a good student");
    i=i+1;
}while(i<=4);
```

Body of loop

i=1 — printf ✓ — 5<=4

i=2 — printf ✓

3<=4 i=3 — printf ✓

4<=4 i=4 — printf ✓

↑ post execution Condition test

Slide



do-while loop

Because condition is tested after execution of body of loop even if condition is Not true at least 1 time body of loop will be executed.

Slide

#Q Consider the following C program:

```
#include <stdio.h>
int main(){
        int a = 6;
        int b = 0;
        while(a < 10) {
                a = a / 12 + 1;
                a += b;
        }
        printf("%d", a);
        return 0;
}
```

Which one of the following statements is CORRECT?

(A) The program prints 9 as output
(B) The program prints 10 as output
(C) The program gets stuck in an infinite loop
(D) The program prints 6 as output

(6<10)

Second iteration $a = 6/12 + 1 = 0 + 1$

1<10

$a = 1$

[C]

$a = a + b$

$= 1 + 0$

$a = 1$

(1<10)

$a = 1/12 + 1$

$0 + 1 = 1$

$a += b = a = 1 + 0 = 1$

$a += b = a = a + b$

$a -= b = a = a - b$

$a /= b = a = a / b$

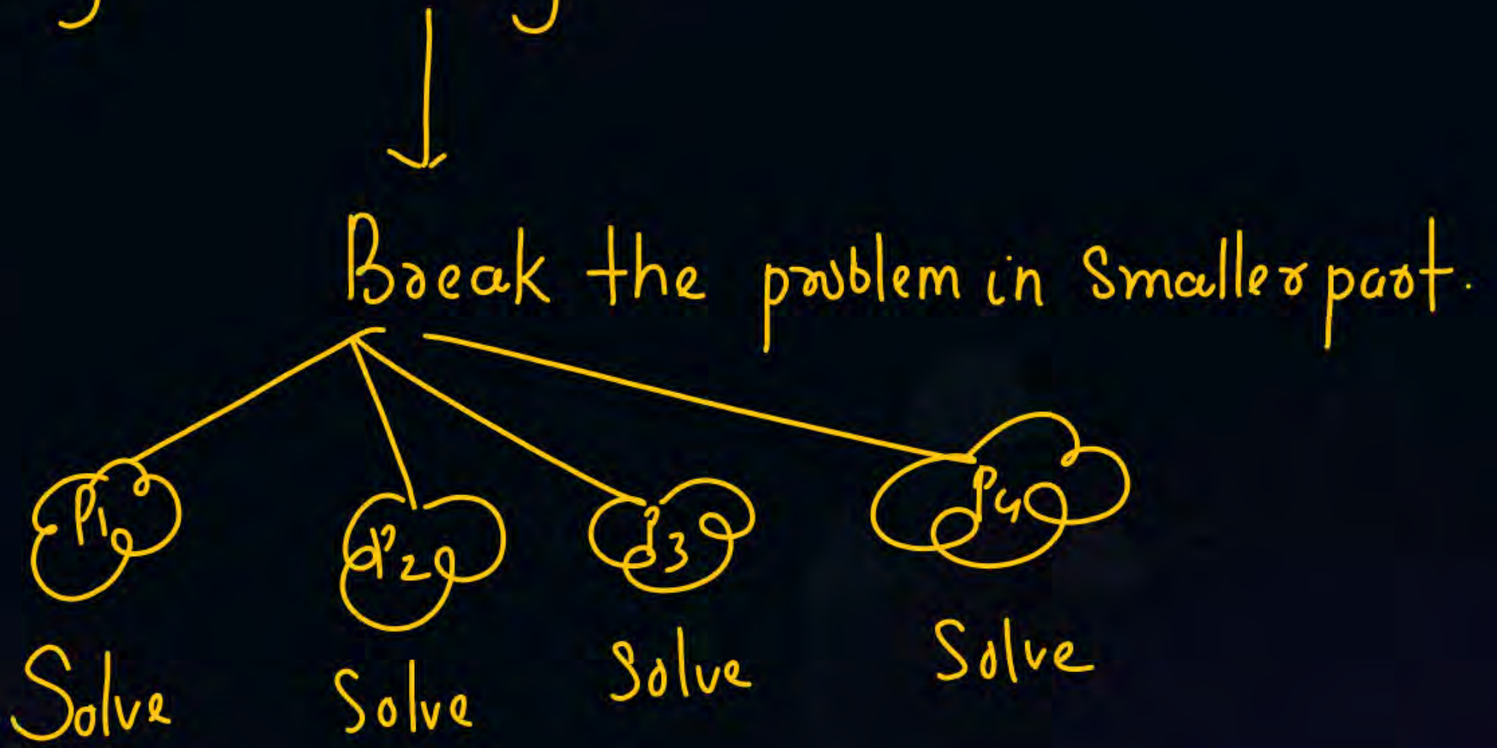
$a *= b = a = a * b$

$a \%= b \Rightarrow a = a \% b$

Shorthand

Function

Management program - manage



Slide



Function

for given problem create Small module dedicated for a task.

- Complexity Recluce
- Reusability

function() - Smaller unit of big program.

Sam program can be used in different places

Slide



Function

1. function declaration
2. function definition
3. function call

example

```
int add(int, int);
```

parameter

return value type

Name of function

Slide



Function

add is a function takes 2 integer parameter and return an integer..

```
int add (int, int);
```

← function prototype

or

```
int add (int a, int b);
```

Slide



Function

function prototype

```
void print();
```

print is a function takes No parameter as input and returns Nothing.



Function

```
#include<stdio.h>
int add(int,int);
int main(){
    int a=10, b=20, Sum;
    Sum = add(a,b);
    printf("%d", Sum);
    return 0;
}
```

declaration

Sum = 30

function call or Invocation

output = 30

```
int add(int c, int d){
    int i;
    i = c+d; //i = 10+20
    return i; = 30
}
```

c 10

d 20

return 30

definition



Function

1. When function is called control transfer to called function execute the function & returns back.
2. If function defined to return some value then return statement used with value to be return.
3. return causes termination of function.
4. if we pass argument to a function: add(a,b)
   ↑ argument.
   the value of argument is copied to parameter in function

Slide



Function

definition `int add(int c, int d)`

parameter

this process of copying argument value to parameter is called ("Call by value")

Actual parameter is (a, b) add(a,b)

formal parameter (c, d)

Slide



Function

```
# includ<stdio.h>
int Sub(int c, intd){
    int i;
    i = c- d;
    return i;
}
int main(){
    int a=10, 6=20
    int diff;
    diff = Sub(b,a)
    printf("%d", diff)
    return 0;
}
```

*

if you want to skip declaration then you can skip and defind the funchon befor its use.

Slide



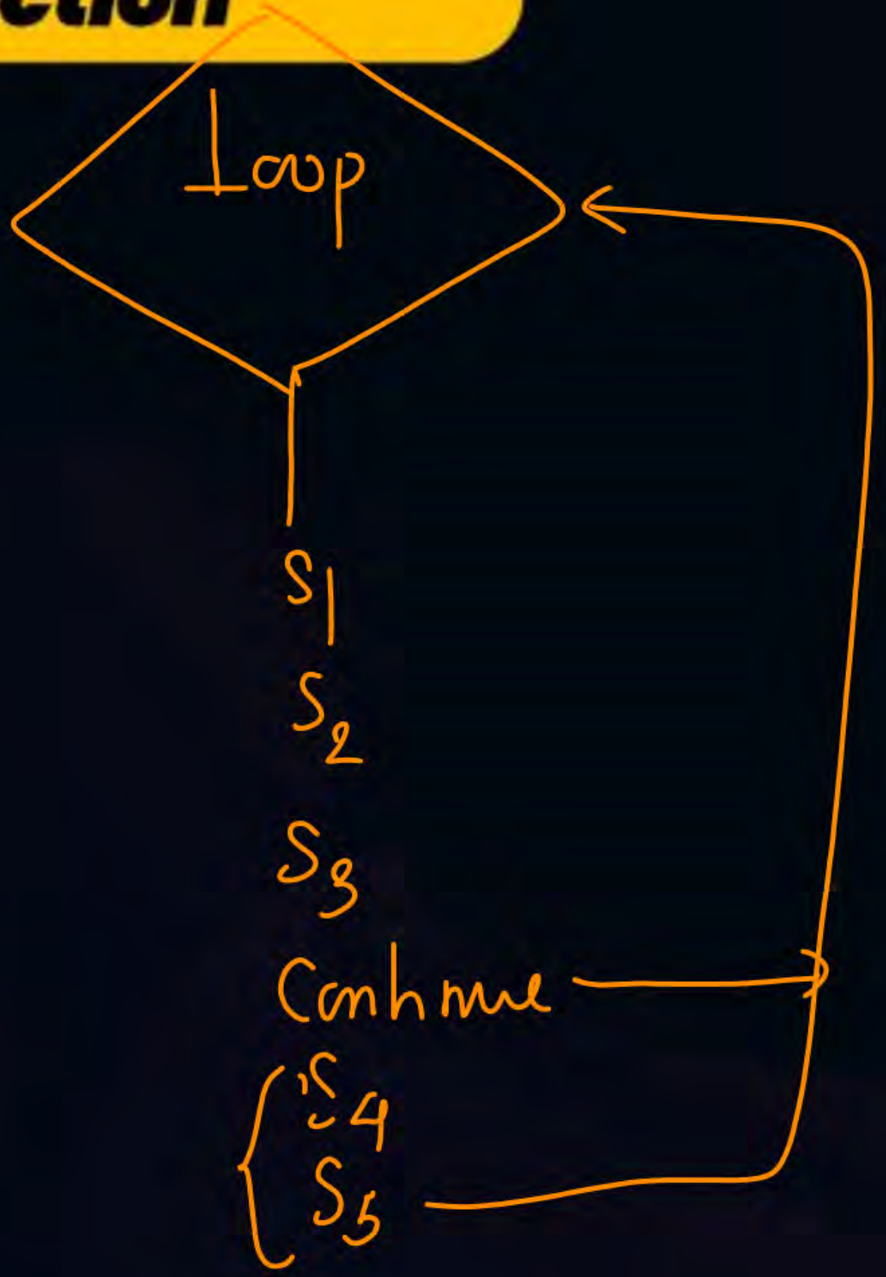
Function

if prototype is Not defined (function declaration is Not given) and directly definition is given after main function then it is assumed by default function return integer.

Slide



Function



Slide

#Q

No. of times '*' will be printed by the following C code is

```
#include<stdio.h>
void foo(int x)
{
        switch(x){
                case 1: printf("*");
                case 2: printf("*");
                case 3: printf("*");
                default: printf("*");
        }
}
int main()
{
        foo(2.5);
}
```

parameter

int x = 2.5;

x = 2

argument

function call

Number of star printed is ?

(A) 2

(B) 3

(C) 4

(D) 5

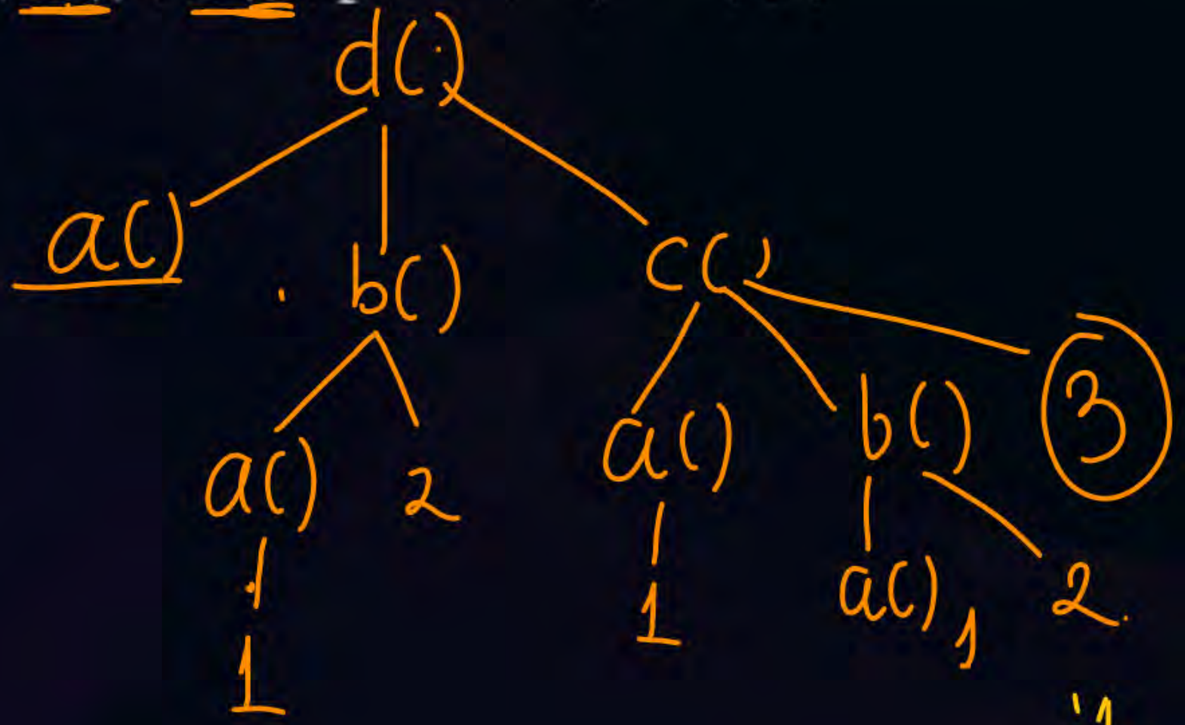
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1 1 2 1 1 2 3 4 before each call function alor

#Q Consider the following program

```
#include <stdio.h>
void a(){printf("1");}
void b(){a();printf("2");}
void c(){a();b();printf("3");}
void d(){a();b();c();printf("4");}

int main(){
    d();
}
```



1 1 2 1 1 2 3

What is the output of the following program?

(A) 11211234

(B) 11211243

(C) 11212134

(D) 11211324

Tomorrow will repeat



2 mins Summary

Topic

Topic

Topic

Topic

Topic

function declaration

function definition

argument & parameter

Slide

THANK - YOU

